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Cotton Leaf Disease Detection using Convolutional Neural Networks (CNN)

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Abstract*: Deep learning is a subset of artificial intelligence. It's a form of artificial intelligence and machine learning that attempts to simulate the way humans pick up specific types of information. The goal of this project is to create a deep learning model based on convolutional neural networks that can distinguish between healthy and diseased leaves. Due to its useful features in learner autonomy and extraction of features, it has drawn a great deal of attention in past years from researchers and industry professionals alike. Images of healthy and rotting leaves are included in the dataset. It is widely used in fields such as computational linguistics, voice processing, image processing, and video processing. It has also become a center for studies on agricultural plant protection, such as the detection of plant diseases and the assessment of pest ranges. This study has also discussed about some of the problems and issues that are currently being faced and need to be addressed. Library packages such as KERAS, MATPLOTLIB, NUMPY, and OPENCV have been utilized here.

Keywords- Dead leaves, Artificial Intelligence, Color Analysis in images, Feature Extraction, Deep Learning

I. INTRODUCTION

When plant diseases spread, they not only threaten the world's ability to produce food, but also have devastating effects on the economies of smallholder farmers.[14] Forecasts of crop loss of much more over 50percentage points due to diseases and pests are common, and more over 80percent of total of farmland production in the emerging world is generated by smallholder farmers.[15] The large percentage of the world's hungry population consists of smallholder farmers, making them especially susceptible to pathogen-caused disruptions in the food supply. To combat crop failure due to disease,

numerous programs have been developed.[16] Over the past decade, integrated pest management strategies have largely superseded the use of insecticides in the pest control process. No matter the strategy, early and correct diagnosis of a disease is essential to effective disease management. [17]Local plant clinics and agricultural extension services have historically played a role in disease identification. Recently, the accessibility of online assets for disease detection has aided these initiatives by capitalising on the rise in Internet usage around the world.[18] Taking advantage of the unprecedented widespread adoption of cell phone technology, a plethora of cell phone tools have emerged in recent years. Diagnosing plant diseases visually is laborious, inaccurate, and impractical outside of specific environments.

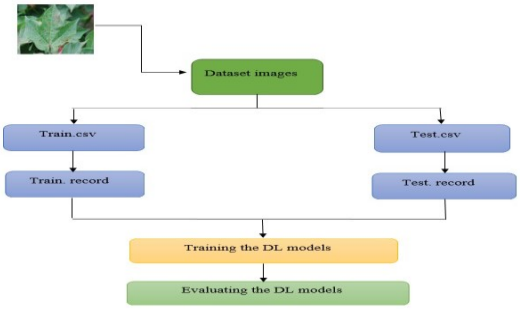

Fig 1: Flowchart of Training Pipeline

The constant monitoring system may also help in the fight against disease and infection. classification, with plant illness being the most significant discovery. Agriculture-related academic discipline. Researchers have explored

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many various operating prototype and deep learning stages, as well as many different possible alternatives. The focus of this piece is on the effects of ICT on information. Image collection, disease identification, brown spot, hispid, leaves attack, and wellness evaluation using deep learning: these are the four classes of data we'll be gathering, processing, and training on. In this article, we examine the various approaches taken when classifying plant diseases. The CNN-based network can be trained to identify plant diseases with a large sample of images of both diseased and healthy plants; once trained, the model can predict plant diseases based on images of plants.

II. LITERATURE SURVEY

The proposed method has the potential to be extremely useful in diagnosing stemcentric diseases in jute plants. figuring out which parts of a plant's leaf are diseased and what kinds of diseases those leaves have by feeling their textures Generally speaking, there are four main stages in the process of disease diagnosis: A color transformation structure is applied to the input RGB image, and then the green pixels are removed selectively using a threshold.[1]It was proposed in a paper titled "Identification and Categorization of Leaf Using Artificial Neural Network" that sick leaf images could be used to diagnose plant diseases. An ANN is trained to make this distinction between healthy and diseased samples by carefully choosing feature values.

Approximately 80% of predictions are correct using the ANN model.[2] Useful chunks are generated by computing statistics. The collected features are then fed into a classifier, which assigns the disease to one of several predetermined categories.[3]Plant leaf diseases are discovered and categorized using Deep Neural Networks. In this paper, we use a Deep Learning approach to discover plant diseases and categorize them by inspecting leaf images.In this paper, we use the convolutional neural network YOLOv3, which has been trained on ResNet18 models to extract leaves from photos.[4]Cotton leaf illnesses are classified in this paper using deep learning techniques. To accomplish this feat of image categorization, this paper employs the LeNet structure, which is a convolutional community architecture.

Image Recognition with Deep Residual (4th):[5] In this paper, we present a more comprehensive residual learning framework than was previously available. In these models, the layers are reformulated so that the focus is on studying residual features and their relationship to the layers' inputs rather than on analysing other, untapped features. [7]Large-scale plant categories benefit from the use of deep neural networks. Flowers classification using deep learning strategies is discussed, as is its use for largescale biodiversity tracking, and how the plant classification using convolution layers society configurations like 50 iterations offers better accuracy than other plant classification applications. [8]Segmentation and soft computing techniques for identifying plant leaf diseases: In this paper, we use a predefined set of rules (a segmentation method) to analyse images in order to classify and identify leaf diseases.

The purpose of this paper is to present a probabilistic neural network-based algorithm for classifying plants based on their leaves. The eleven characters from the leaf are collected. CNN has been trained to differentiate between 32 different types of flowers and 1800 different types of leaves, resulting in an accuracy of more than 90%. These models are more precise than a generic CNN version and require less training time to implement.[9]Several frameworks are discussed in this paper, all of which make use of numerous innovations to increase participant representation on the board. While CNN has found widespread application in issues involving visual facts, with excellent standard overall performance achieved, its implementation guidelines have the most significant shortcomings. When information is routed through CNN's pooling layers, it loses some of it, and because of its inability to express perspective invariance, a new class of neural network depictions called the tablet community has emerged as a promising alternative.

III. EXISTING SYSTEM

In ANN models, the complexity of the human brain has been greatly reduced. The computational units in an ANN are called artificial neurons and they are designed to be functionally equivalent to their biological counterparts in the nervous system. There are typically three main parts to an ANN model: input, hidden, and output. A signal travels from each nth layer neuron to its (n+1)th-layer counterparts. Every link is given a certain amount of importance. Multiplying each input by its relative importance yields the final result. When constructing an ANN, the output is fed into an activation function. The ANN has the potential to aid in the resolution of numerous engineering and scientific issues. Thus, the ANN can be used for many different things [3] including jpeg, statistical models, mathematical models, stock market estimation, diagnostics, and signal

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processing.

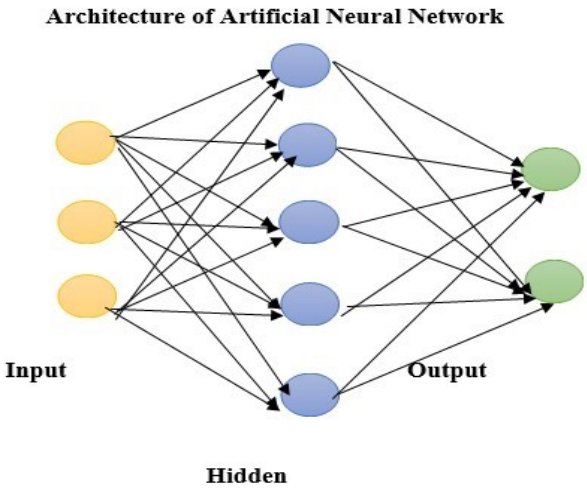

Fig 2: Basic Diagram For ANN

There are a number of issues with ANN, including the fact that: •Artificial Neural Nets require a large amount of computational power; • Neural Networks models are difficult to explain; • Neural Networks need large amounts of data for training; • Data planning for model of neural networks must be done meticulously.

IV. PROPOSED SYSTEM

The most important economic drivers is the agricultural sector. About half of India's workforce is employed in the agricultural sector. Photons, rice, grains, ingredients, and spice products are just some of the many ingredients that India is famous for producing in abundance. The prosperity of farmers is tied to the value of the goods they produce, which in turn is determined by the health of their plants and the quantity of food they harvest. The ability to identify plant diseases is, therefore, crucial in the agricultural sector. As a result of the widespread prevalence of plant-killing pathogens, farmers' efforts to preserve biodiversity are frequently thwarted. Using a fully automated disease detection method is helpful for discovering plant diseases at their earliest stages of development. Plant diseases manifest themselves in various ways, with symptoms visible in the leaves, stems, roots, etc. The process of manually identifying plant diseases from images of their leaves is laborious. In order to automate the process of detection of disease using images of leaves, new computational methods must be developed.

Structures and Reference Materials:

Matplotlib and Keras (the Tensor Flow Library Interface) (For plotting images and graphs)

- Python package (For dealing with multi-dimensional arrays)
- OpenCV (For dealing with images)

Some procedures must be carried out to determine if whether leaf is fresh or has been infected with a disease. I.e., Data Preparation, Feature Extraction, Classifier Training, and Classification. In image processing, "preprocessing" means reducing the overall size of all the images to the same level. The next step is to use an algorithm to pull features from a pre-processed image. Four categories of disorders were considered when collecting the images. Less visuals were available. Since additional images are needed for deep learning-based methods, we have supplemented the data with
Matplotlib. This one-time method generates ten mirrored copies of the original image. Therefore, a large amount of data is generated that can be used to refine the model. Our team has made use of many different libraries for data science, such as Keras, vector flow, OpenCV, matlab, and Numpy.

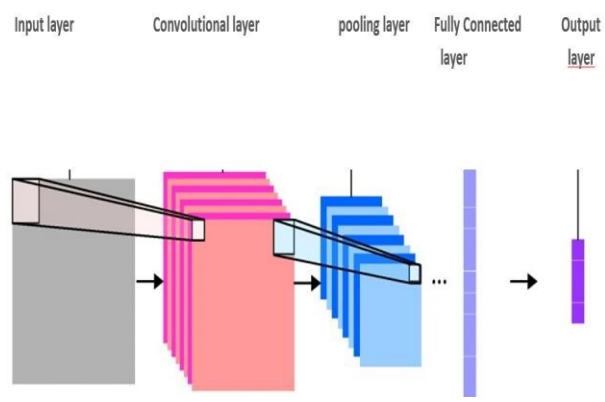

Fig 3: Layers Of CNN

To build the Keras model, we used sequential modelling. There are 2 conv2D layers in the model's architecture. The information can be found on the coLab server. The path variable will sequentially examine each photo along the way. Each picture is decoded using the OpenCV library. After that, dyadic image processing is used to reduce the size of the photos. Every photo's Class is extracted from the filenames and stored in a label variable, which also contains other meta data about the images. The model is trained by first transforming the label lists and input data

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into Numpy arrays. The data train is 75% longer than the exam, which is 25% longer. During augmented data gleaned from actual running instances. The classifier is given an image with 128x128x3 pixels and information about 4 classes.

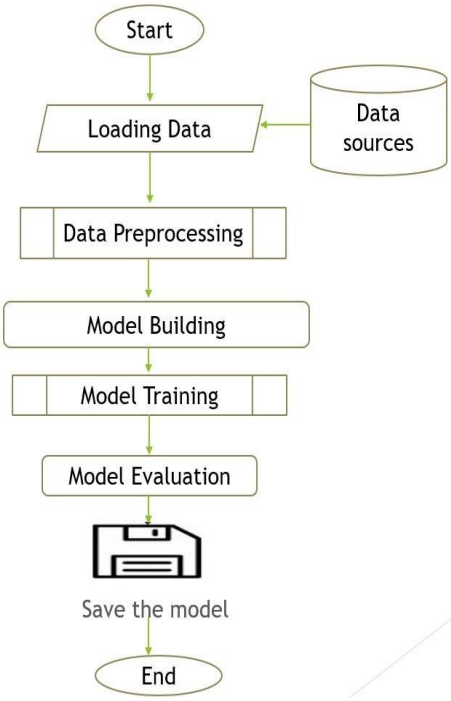

Fig 4: CNN Model Flow Chart

The Step Before Taking an Image:

Digital images undergo pre-processing, or image processing, with the help of computer algorithms. Finding the plant requires applying a specific algorithm to the visual analysis process. For image

	Training data	Validation
No of fresh cotton leaf images	427	80
No of diseased cotton leaf	288	55
Total no of images	715	135

matplotlib, and NumPy. To build the Keras model, we used sequential modelling. There are 2 conv2D layers in the model's architecture. The information

training, the optimizer also receives

The first step is to obtain the data from a freely available database.

analysis and detection, we take a similar approach and employ the same algorithm. We can't use an algorithm based on a blurry image because picture quality is crucial to this procedure.

Image editing, also known as image enhancement, is the process of making adjustments to digital photographs in order to make the final product look better on screen or be more amenable to further processing. Any of the ones that follow can be used to enhance an image:
Heterogeneity correction, Noise can be reduced with the help of filters, Stretch, decorrelation, unsharp mask filtering, etc.

Image segmentation:

The term "image segmentation" refers to the process of dividing a digital image into smaller individual pictures (sets of pixels, also known as image segments objects). Segmenting an image into its constituent parts and then analysing them individually can greatly facilitate image analysis and identification. It's not uncommon to find a common thread running through each of the subgenres, particularly in terms of texture, and intensity.

DATASET:

The subsequent step is to acquire the input test image, then perform any necessary pre-processing, and finally convert the image to array format for analysis. After selecting a database, it is properly pre-processed, partitioned, and renamed before being placed in its respective directories. CNN organises the data after properly training the model. Once the picture and the evidence of strong have been correlated, the results can be displayed. The software will show the ailment and the treatment options if the plant is afflicted.

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can be accessed on the server in CoLab. The path variable will sequentially examine each photo along the way. Each picture is decoded using the OpenCV library.

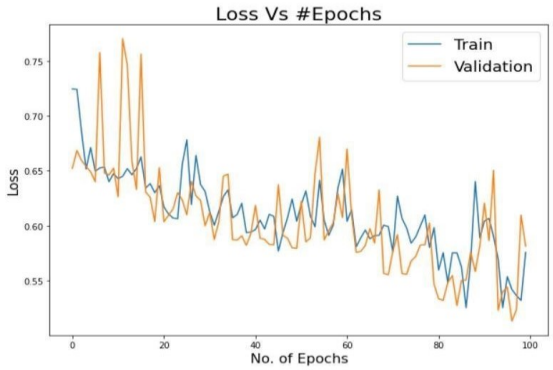


Fig 5: ANN model graph for loss vs Epochs

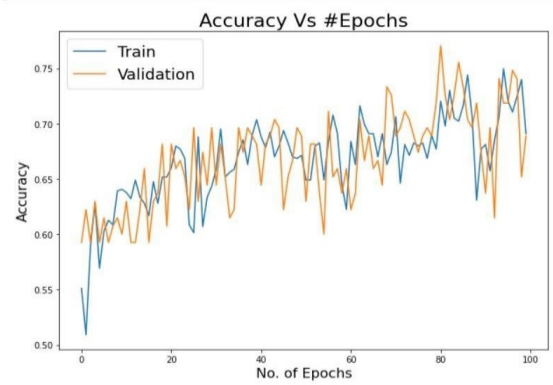


Fig 6: ANN model graph for accuracy vs Epochs

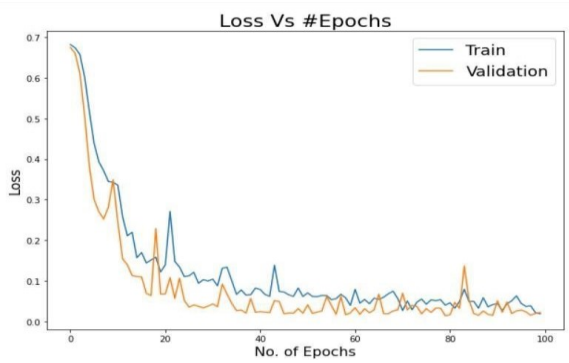


Fig 7: Feature Selection Accuracy level

V. RESULTS

ANN Model Accuracy:

- Training Accuracy after 100 epochs: 66.21%
- Validation Accuracy after 100 epochs: 68%

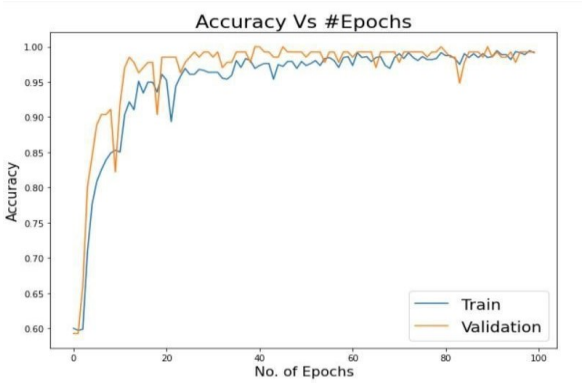


Fig 8: CNN model graph for accuracy vs Epochs

Feature extraction Accuracy level:

Once compared to the current model, the extracting features overall accuracy will be quite low, as shown by the following graph:

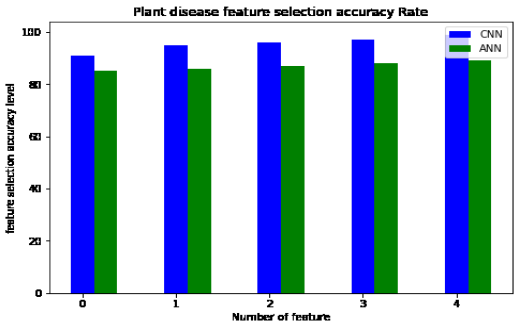
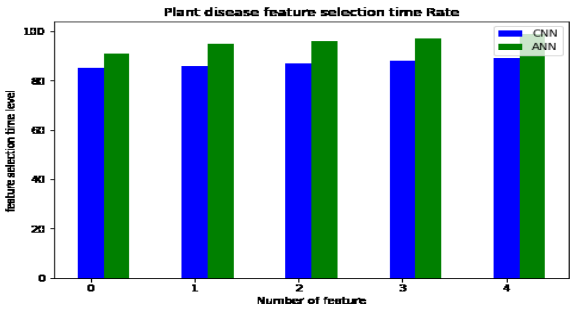


Fig 9:

Fig 9: Feature Selection Accuracy level



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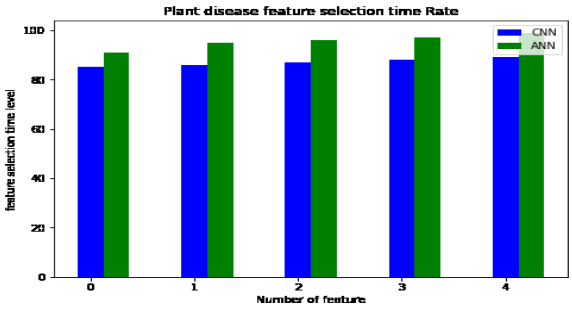

Fig 10: Feature selection Time level

Testing CNN Model for Custom Input:

Fig 11: Prediction of Diseased cotton leaf

VI. CONCLUSION

The training set can be implanted into the Internet of things like the Arduino and Raspberry Pi, integrated with drones, and deployed into the fields; the CNN Model outperformed the ANN Model in terms of accuracy.

The aim of this study is to develop a system for identifying diseases in cotton crop leaves. The cotton plant's leaves are its primary infection vector. No matter how much a cotton leaf changes in hue or feel, the human eye can't tell the difference. Therefore, a replacement method for plant detection is required.

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